ANSWER KEY & MARKING SCHEME SET 1

Chapter 15: **The Making of a Scientist (Footprints Without Feet)** | Class 10 English | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Robert W. Peterson	Robert W. Peterson wrote this biographical piece.
2	**(d)** pupa	Gold spots on the pupa.
3	**(d)** inspiring and informative	Biography of achievement.
4	**(a)** Richard H. Ebright	It follows Richard Ebright.
5	**(c)** a hormone needed for full development	A hormone.
6	**(a)** how a scientist is shaped	Process of becoming.
7	**(a)** butterflies, rocks, fossils and coins	He was a keen collector.
8	**(a)** Harvard	At Harvard.
9	**(b)** invent a telescope for the army	Not mentioned.
10	**(d)** the viceroy	His science-fair project asked whether the viceroy mimics the monarch.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Richard Ebright is an American scientist (1) whose early life and work are described in the chapter (1).	1+1
12	He tested whether the viceroy butterfly imitates the monarch (1) to escape being eaten by predators (1).	1+1
13	Curiosity and a love of observing nature (1) and patience and hard work in experiments (1).	1+1
14	He collected butterflies (1) and also rocks, fossils and coins (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	As a child Richard loved to collect things: butterflies, rocks, fossils and coins. He completed a collection of the local butterflies at an early age. His curiosity about nature began early.	1+1+1
16	He started with collecting and tagging. Then he did experiments on the viceroy and the monarch pupa. Later he studied DNA and cell life at Harvard.	1+1+1
17	Richard's mother said that she would write to Dr Urquhart. 'I' becomes 'she' and 'will' becomes 'would'.	1+1+1
	OR: The book told the story of monarch migration. It aroused his curiosity and led him to tag monarchs. It was the beginning of his research.	1+1+1
18	He is curious, observant, patient and persistent. He asks questions and tests them by experiment. He does not give up easily.	1+1+1
	OR: Richard collected butterflies and tagged monarchs. Both verbs are in the simple past.	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	A curious collector of butterflies and rocks (1). Encouraged by his mother (1). Tagging monarchs and experimenting (1). Persistent and hardworking (1). Reached Harvard and DNA research (1).	2+2+1
	OR: Address and date (1). Salutation (1). Early life (1). Achievements (1). Closing (1).	2+2+1
20	She noticed his interest (1). Gave him the book on monarchs (1). Wrote to Dr Urquhart (1). Supported his projects (1). Her role shows a parent's importance (1).	2+2+1
	OR: Follow your interests (1). Be curious (1). Work hard (1). Use support from parents and teachers (1). Never stop learning (1).	2+2+1

Mistakes to avoid:

- Mixing the scientist's name: the chapter is on **Richard H. Ebright**, an American biologist, not on an Indian scientist.
- Saying he became a scientist only because of school; his **mother's support** (trips, telescopes, microscopes) and his collections drove his early curiosity.
- Forgetting the collections: from kindergarten he collected **butterflies**, then rocks, fossils and coins, and became an eager astronomer.

- Missing the turning point: a **book** on butterflies (and a later project on monarch butterflies) shaped his career; the chapter asks how a book changed his life.
- Calling him a famous scientist as a child; he was a shy boy who could not play team sports and so *collected things*.
- Spelling slips: *Ebright, scientist, curiosity, butterfly, telescope*; and tense mixing.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 15: **The Making of a Scientist (Footprints Without Feet)** | Class 10 English | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** curiosity and persistence	Love of discovery.
2	**(b)** His mother told him to read that book.	Imperative changes.
3	**(c)** The Travels of Monarch X	A book on monarch butterflies' migration.
4	**(c)** marking butterflies to follow their movement	Tagging.
5	**(b)** indifferent	Indifferent.
6	**(c)** the migration of monarch butterflies	Monarch migration.
7	**(b)** seasonal movement from one region to another	Migration.
8	**(d)** determined	Persistent = keeps trying.
9	**(a)** take part in tagging monarch butterflies	Tagging was a research project.
10	**(d)** he completed projects on his own initiative	He worked eagerly.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The spots were not just decoration (1); they produce a hormone needed for the butterfly's full development (1).	1+1
12	Early interests may shape one's career (1) if encouraged by parents and teachers (1).	1+1
13	She gave him 'The Travels of Monarch X' (1) and wrote to Dr Urquhart so that he could join the tagging of monarchs (1).	1+1
14	They gave him books and time (1) and supported his projects without forcing him (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	She noticed his interest and gave him the right book. She contacted Dr Urquhart for tagging. Her support gave him confidence and opportunities.	1+1+1
16	They give books, tools and encouragement. They let the child explore. They connect the child to experts as Richard's mother did.	1+1+1
17	Monarchs show long migration. A hormone controls growth. Richard did an experiment on the viceroy.	1+1+1
	OR: Richard wondered if the viceroy imitates the monarch. He designed an experiment to test whether predators avoid the viceroy because it looks like the monarch. The project won recognition at a science fair.	1+1+1
18	Tagging is putting a small mark or label on an animal. Scientists then follow the animal's travels when it is found again. Richard tagged monarch butterflies.	1+1+1
	OR: A hobby is done for pleasure while research seeks new knowledge through method. Richard turned his hobby into research by asking questions. This is how a scientist is made.	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Collecting local butterflies (1). The book on Monarch X (1). Tagging with Dr Urquhart (1). Viceroy and pupa experiments (1). Prizes at science fairs (1).	2+2+1
	OR: Date (1). Excitement (1). Description of the experiment (1). What he learned (1). Plans (1).	2+2+1
20	Childhood interests grow (1). Parents give opportunities (1). Effort and experiments (1). Education at Harvard (1). Conclusion (1).	2+2+1
	OR: Child: collector, curious (1). Scientist: researcher, experimenter (1). Both: love of nature (1). Growth through support and effort (1). Conclusion (1).	2+2+1

Mistakes to avoid:

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 15: **The Making of a Scientist (Footprints Without Feet)** | Class 10 English | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** they helped him test and show his ideas	They were stepping stones.
2	**(c)** trace their migration	Tagging shows their travels.
3	**(d)** Butterflies were collected by Richard.	Past passive.
4	**(a)** observe and classify carefully	Collecting trains observation.
5	**(b)** they gave him books, support and freedom to explore	Parents were supportive.
6	**(c)** chemical messenger in the body	Hormone.
7	**(c)** past perfect	had + V3.
8	**(b)** imitates the monarch	Mimicry of the monarch.
9	**(b)** that the gold spots were not merely decoration	They had a function.
10	**(d)** curiosity, encouragement and hard work make a scientist	Theme.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It gave data on where butterflies travelled (1) and helped understand migration patterns (1).	1+1
12	Placing small tags on monarch butterflies (1) so that their long migration routes can be traced (1).	1+1
13	At Harvard (1) where he studied how cells read their DNA (1).	1+1
14	A scientist is made by curiosity, support and effort (1) and any child can grow into one (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Many thought the spots were only ornament. Richard experimented and found that they give out a hormone needed for the development of the butterfly. This was a real scientific discovery.	1+1+1
16	Collecting taught him to observe and classify. It trained his memory and patience. It laid the base for scientific work.	1+1+1
17	Real research needs patience and repeated trials. Richard worked for long hours to test his ideas. Mistakes are part of learning.	1+1+1
	OR: They gave Richard a chance to test and present his ideas. Success encouraged him to continue. They are shown as a step in the making of a scientist.	1+1+1
18	Science begins with questions about the world. Curiosity makes one observe and experiment. Richard's questions about butterflies led to discoveries.	1+1+1
	OR: It shows how a boy's simple hobby became a career. It shows the value of support and effort. Any student can take a similar path.	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Viceroy imitates monarch (1). Test with predators (1). Gold spots produce hormone (1). He proved it by experiment (1). These won prizes (1).	2+2+1
	OR: What tagging is (1). How it is done (1). What data it gives (1). Role of Dr Urquhart (1). Importance to science (1).	2+2+1
20	Opening (1). What curiosity is (1). Richard's example (1). Benefits for learning (1). Conclusion (1).	2+2+1
	OR: Greeting (1). Story of Richard (1). Steps: curiosity, hard work (1). Role of support (1). Closing appeal (1).	2+2+1

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